Case Study: Drive a Human-Machine Learning Loop by Enabling ML Literacy Among Business Analysts

2 October 2024 - ID G00783057 - 7 min read

By: Finance Research Team

Initiatives: Technologies and Markets

Decision makers hesitate to use machine learning in planning because they are unfamiliar with how to use its outputs and insights. To drive a stronger human-machine learning loop in planning, FP&A leaders should improve analyst-level ML literacy and translation skills.

- Company Name: Huntsman
- Industry: Chemicals
- Headquarters Location: Texas, U.S.A.
- Revenue: $5 Billion to $10 Billion
- Employees: 9,000+ (February 2023)

Case Overview

Problem

- Despite the transformational potential of machine learning (ML) in planning, many decision makers in the business are unfamiliar with how to use its outputs because they don't always fully understand this new technology. Instead they prefer familiar, traditional planning methods.
- Upskilling only decision makers on using and interpreting ML outputs is insufficient because this concentrates ML knowledge in a few stakeholders. This concentration creates talent bottlenecks that make it impossible to address all of the organization's ML planning needs.

This research note is restricted to the personal use of .

Action

- **Knowledge Management Use-Case Library:** Huntsman created an ML use-case library that centrally stores information from the entire organization to improve knowledge management and collaboration.
- **Last Mile Translation:** The company selected ML model translators from analysts with the capabilities to perform confirmatory data analysis and explain model outputs to stakeholders.
- **Human-Machine Friendly Competition:** Huntsman proved ML efficacy in planning to the entire organization by setting up high-visibility forecast accuracy competitions between analysts and ML.

Results

- The number of staff able to use ML quadrupled over the course of a year without adding any FTEs, and ML use cases increased from just five to 65 over the same period.
- Decision makers have embraced ML in planning, now viewing it as the default method for processes with existing ML use cases.

Problem

To navigate economic volatility, many organizations are turning to ML to increase their financial planning agility. However, ML implementations are commonly undercut by decision makers not fully understanding this new technology. This impedes the human-machine learning loop, which is a proven driver of planning agility. Decision makers are unfamiliar with how to use the outputs of these models during financial planning. Instead of ML, they revert to traditional planning methods they know and trust.

Organizations often believe they can address this unfamiliarity with ML by training decision makers on how it works. However, this approach is insufficient and leads to a limited group of individuals with ML knowledge. This concentration of talent means any ML-related questions or needs must pass through these decision makers. The resulting bottleneck delays addressing these ML needs and minimizes the potential impact of ML in planning.

This research note is restricted to the personal use of .

To address decision makers' unfamiliarity with ML, Huntsman improved ML application literacy among a broad group of business analysts from all functions. Multiplying the number of individuals who can address the decision makers' ML needs increased broader ML knowledge. The idea sharing between analysts from all areas of the business also created higher model quality that strengthened the organization's trust in ML and enabled FP&A to increase ML use in financial planning.

Knowledge Management Use-Case Library

Many finance functions struggle to share existing ML implementation knowledge with analysts because that information exists in many parts of the organization. This siloed information prevents analysts from building off an existing model and limits their visibility into the impact of ML on planning processes.

Recognizing the issue of siloed ML knowledge, Huntsman created a use-case library to track information on ML use cases developed by their analysts from across the organization. This library tracks each use case's area of application, the problem it solves, technology details, results and who was involved in developing the model.

Figure 1: Partial Snapshot of Huntsman's ML Use-Case Library

Partial Snapshot of Huntsman's ML Use-Case Library

Use Case	Planning Problem	Tech Used	Results	Model Author
Demand Planning	Demand planning by customer/material		Cross-functional alignment	Business unit analyst
AP Forecast	Statistical forecast for a volatile data set		Greater data visibility	Finance analyst
AR Forecast	Accounts receivable financial forecast		Higher forecast accuracy	Finance analyst
Inventory Forecast	Operational forecast for inventory		Enhance inventory levels	Supply chain analyst

Source: Adapted From Huntsman
783057_C

HUNTSMAN

Gartner

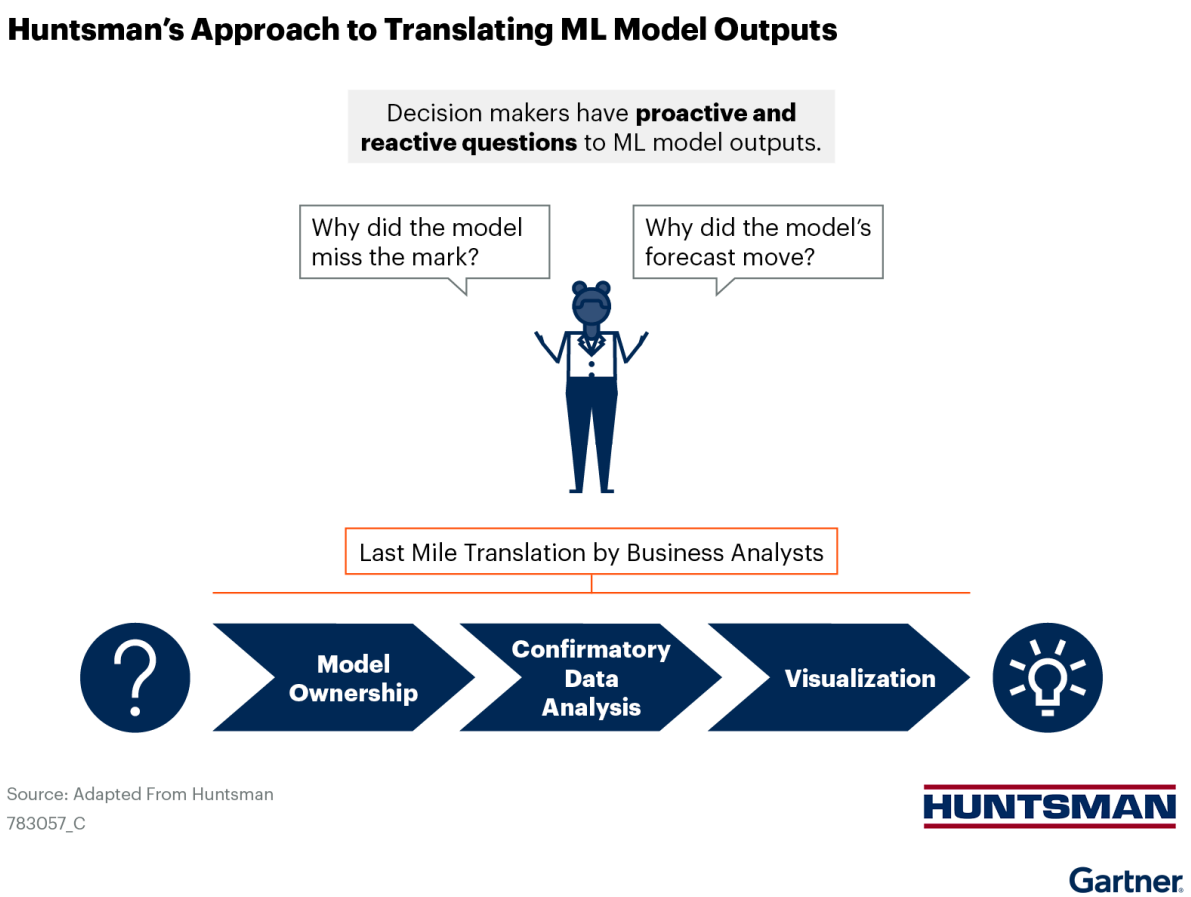
Tracking this information in a centralized library, Huntsman enabled analysts to identify opportunities to collaborate with each other and aggregate individual use cases to address larger planning processes.

This research note is restricted to the personal use of .

For example, the library positioned analysts to improve working capital forecasting by combining use cases for the accounts payable, accounts receivable and inventory forecasting processes. The library also makes it easier for analysts to adapt ML models of existing use cases, even those from a different part of the organization, and share experiences while working on similar projects. Tracking the results of each use case makes it easy for analysts to understand use-case benefits and for finance leaders to see the collective value of ML in planning.

Last Mile Translation

Figure 2: Huntsman's Approach to Translating ML Model Outputs



Increasing the number of ML models did not, however, increase decision makers' use of ML outputs for planning. Unfamiliar with how the technology works, decision makers at Huntsman still questioned why a forecast drastically changed or why a model missed the mark. They sought answers to support their own decision-making comfort and to feel equipped to explain ML-based decisions to upward stakeholders. Without satisfactory answers to these questions, decision makers still resisted using model outputs to inform planning.

This research note is restricted to the personal use of .

Cognizant of the need to fill this last-mile gap in ML implementation, Huntsman taught its analysts to translate the outputs of models to decision makers. These so-called translators lead to greater trust and adoption of ML in planning. The group of translators included analysts from across the organization, creating a broad base of resources to address all decision-maker needs. Huntsman's "last-mile translation" consists of three main components: model ownership, confirmatory data analysis and visualizations.

Model ownership directs decision makers' questions to the analysts with the necessary process and technical knowledge. Those analysts become the owners of that model and their expertise ensures they'll have sufficient knowledge to understand the model and explain it to decision makers.

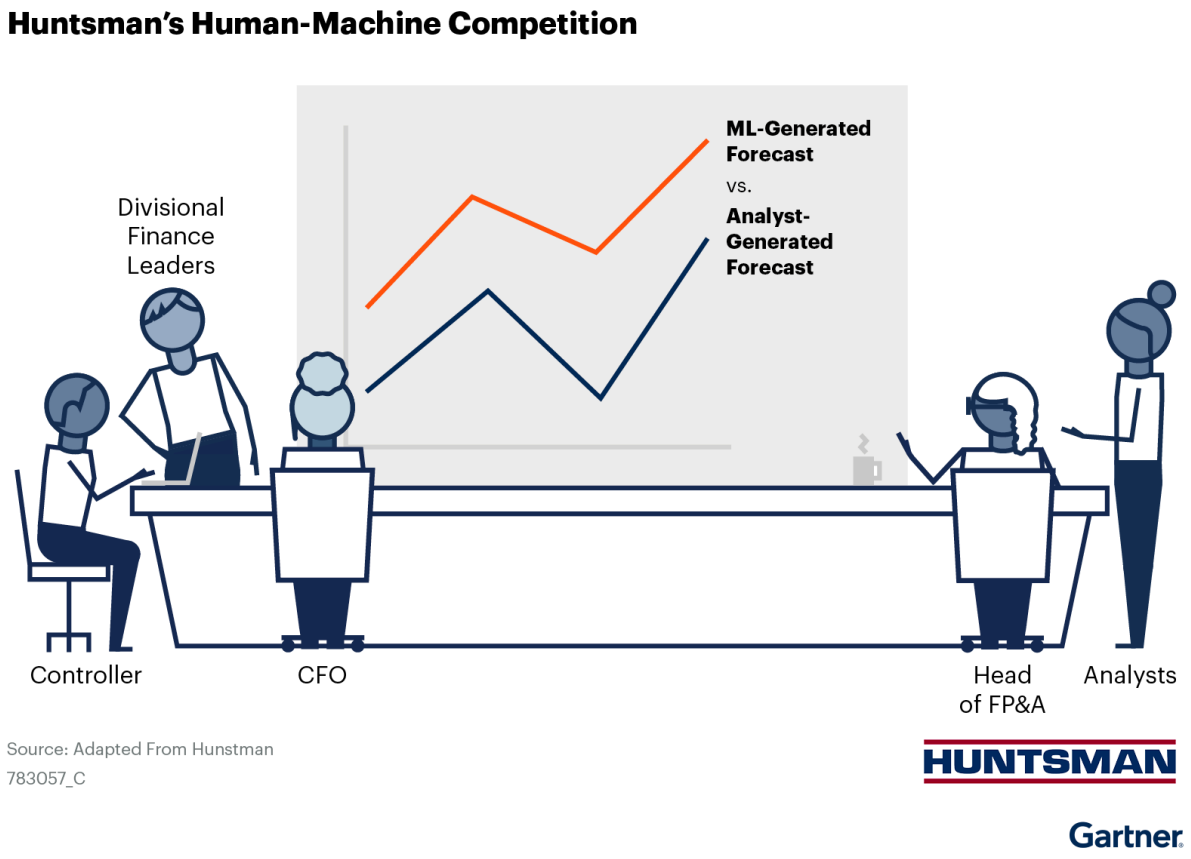
Huntsman's translators must have knowledge of the data underlying the model they take ownership of. This knowledge enables them to perform *confirmatory data analysis* to understand why a model produces a given output.

Finally, translators *visualize* the explanations of model outputs to decision makers to make it easy for them to understand the output and next best actions. The translators build trust in the tool by visualizing model outputs and how the model derived that output.

Human-Machine Friendly Competition

This research note is restricted to the personal use of .

Figure 3: Huntsman's Human-Machine Competition



To make the benefits of ML clear to the organization and increase the adoption of ML models in planning, Huntsman decided to showcase the impact of ML across the firm. To draw maximal attention, Huntsman ran a competition between a human-generated forecast based on traditional methods and one generated by an ML model. The competition tested whether the analyst or the ML model generates a more accurate forecast and measured the manual effort required for each.

Huntsman presented the competition to an audience of key finance and business leaders, such as the CFO, head of FP&A, the controller and divisional finance leaders. Running the competition in front of senior finance leaders generated a lot of interest within the finance organization. Senior leader involvement also created high visibility across the organization and drew employees from many parts of the firm.

This research note is restricted to the personal use of .

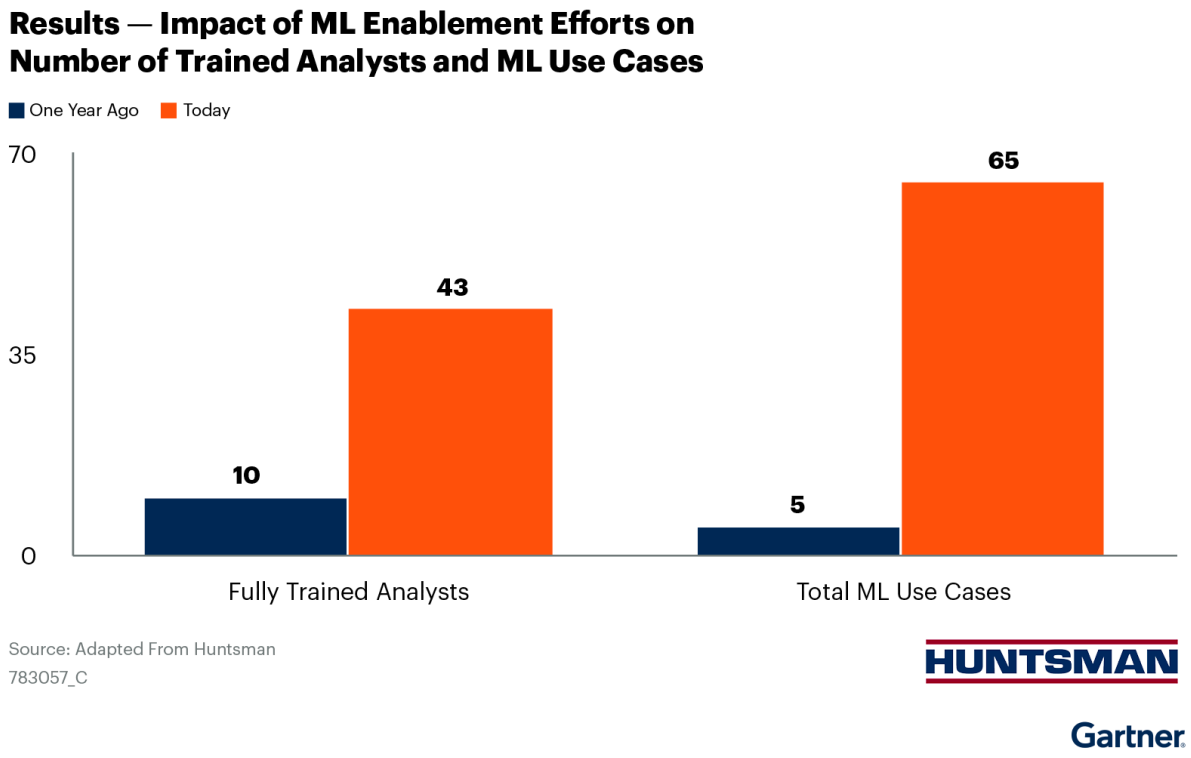
Huntsman selected an analyst for the competition who was enthusiastic about using ML in planning to ensure there would be no resentment regardless of the outcome. Unsurprisingly, the ML model proved to be both more accurate and more efficient than traditional planning methods. By positioning the contest as a friendly competition, Huntsman got both the observing analysts and decision makers excited about the technology. They could study live how the ML-generated forecast led to better results and learned from the analyst competitor about their experiences in using human versus ML-generated insights.

The human-machine friendly competition was a fun way of lessening stakeholders' defensiveness toward using ML in planning and getting the whole organization engaged.

Results

Huntsman's efforts to enable analysts to use ML in planning led to widespread adoption and success with the new technology. The number of analysts trained in using ML quadrupled over the course of a year, from around 10 to 43. As a result, the number of ML use cases at the organization has increased markedly, from just five a year ago to 65 as of publication.

Figure 4: Results: Greater Integration of ML in Planning



This research note is restricted to the personal use of .

This increased engagement from analysts with ML use in planning also made decision makers more comfortable with the technology, leading to significant benefits to financial planning. As Huntsman's vice president of financial planning and analysis, Nooshin Vaughn, put it, "With volatile market realities, you have to be able to respond quickly. Using ML in planning has enabled us to move faster and with greater confidence."

Recommendations

FP&A leaders who want to get their organizations to engage with using ML in planning should:

- Ensure analysts have access to a robust ML knowledge base by creating a library to track any existing or future ML use cases in planning, and sharing the library with relevant analysts and decision makers.
- Ensure decision makers actually use ML in planning by making those analysts with the most relevant process and technical expertise responsible for answering any decision-maker questions on the model.
- Create broad interest in using ML in planning by hosting a demonstration of ML efficacy in front of key finance stakeholders in the form of a friendly competition and making it visible to the organization.

Evidence

We developed this case study to describe Huntsman's approach to enabling greater use of ML in planning. It is based on extensive interviews with Huntsman's finance and global business service leaders, and reviews of internal documentation provided by the company.

Recommended by the Authors

How Leading Finance Organizations Achieve AI Success

Build Your Own AI-Driven Use Cases in Finance

Accelerate AI Initiatives With a Citizen Data Scientist Skills Program Customized for Finance

Presentation Deck

Download the presentation deck for this case study

This research note is restricted to the personal use of .

Recommended by the Author

Some documents may not be available as part of your current Gartner subscription.

Innovation Insight: AI in Financial Planning Software

Case Study: Driver-Based Machine Learning Planning Foundations

Improve Budgeting and Forecasting Agility With AI/ML Using a Human-Machine Learning Loop

How Leading Finance Organizations Achieve AI Success

Best Practice: FP&A-Led Co-Creation of Decision Support Tools (Valmont)

AI Use-Case Library for Finance

© 2026 Gartner, Inc. and/or its affiliates. All rights reserved. Gartner is a registered trademark of Gartner, Inc. and its affiliates. This publication may not be reproduced or distributed in any form without Gartner's prior written permission. It consists of the opinions of Gartner's Business and Technology Insights Organization, which should not be construed as statements of fact. While the information contained in this publication has been obtained from sources believed to be reliable, Gartner disclaims all warranties as to the accuracy, completeness or adequacy of such information. Although Gartner insights may address legal and financial issues, Gartner does not provide legal or investment advice and its insights should not be construed or used as such. Your access and use of this publication are governed by Gartner's Usage Policy. Gartner prides itself on its reputation for independence and objectivity. Its insights is produced independently by its Business and Technology Insights Organization without input or influence from any third party. For further information, see " Guiding Principles on Independence and Objectivity." Gartner insights may not be used as input into or for the training or development of generative artificial intelligence, machine learning, algorithms, software, or related technologies.

This research note is restricted to the personal use of .